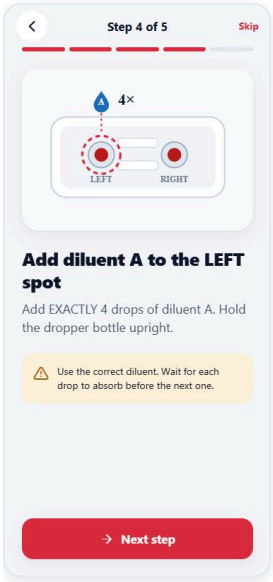
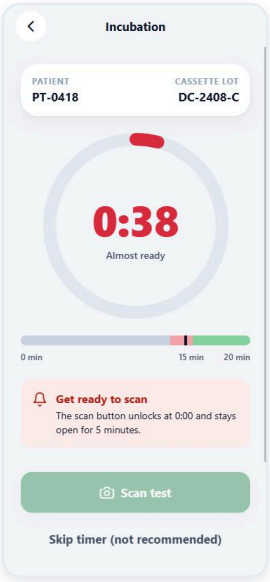
BRAINCheck redesign

A concept prototype of the app flow in the public IFU, rebuilt in React Native / Expo with a clearer interface.

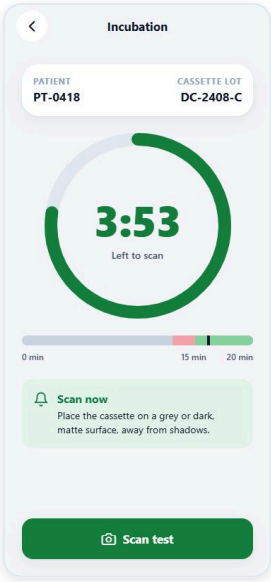
Filip Galetic
Full-stack engineer
github.com/fgaletic · filip@allio.li



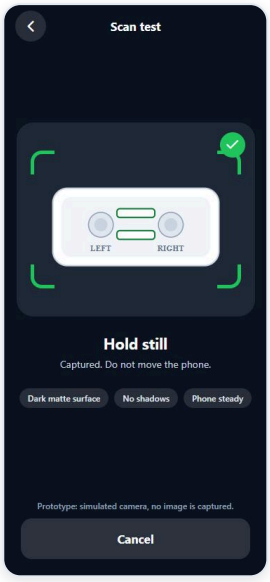
Guided sample steps
One task per screen, exact drop counts called out



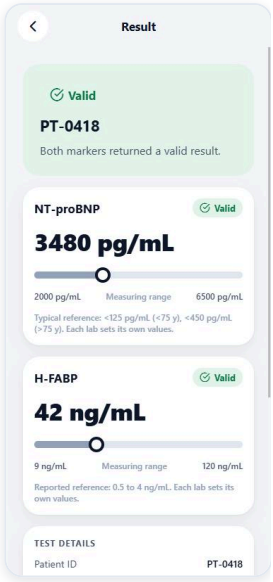
Last 2 minutes
Ring turns red, timeline shows all phases



Scan window
5 minutes, then the cassette is discarded



Scan
Frame turns green and scans on its own (simulated)



Result
Status, value and position in the measuring range

WHAT I EXPLORED, AND WHY

- **Status is never colour alone.** Valid, Incomplete and Invalid are always icon plus text.
- **Glove-friendly.** 58 px buttons and 56 px fields, one task per screen.
- **The timer is the main object.** The IFU has two thresholds and an expiry; the ring, colour and a 0 / 15 / 20 minute timeline make them visible at a glance.
- **No interpretation.** Results show where a value sits in the measuring range, with "<2000" and ">120" shown as the IFU specifies. No normal or abnormal labels.
- **Deviations are visible.** A skipped timer needs a confirmation and is flagged on the result.
- **Scan conditions at the point of need.** Surface, shadows and steadiness are text in the IFU; here they sit on the scan screen.

Try it fast: a prototype control speeds the timer up (30× / 60×) and picks the simulated outcome (valid, out of range, incomplete, invalid).

COVERAGE OF THE IFU

7.1	Four-slide tutorial, skip, "don't show again"
7.2	Lot and patient ID, five guided steps
7.3	15 min timer, red at 13 min, 5 min scan window, timed-out state, skip with confirmation
7.3	Alignment grid, auto-scan (simulated)
7.4	Valid / Invalid / Incomplete, units, limit display, 30-day auto-delete
8	About: support, privacy, terms, regulatory (inert)
+	English and Spanish, switch in About

ENGINEERING

- **Expo 58, React Native 0.88, strict TypeScript.** Runs on web, iOS and Android; checked on web.
- **Pure domain layer** for timer phases, classification, range display and retention, with no UI imports.
- **23 unit tests** and a CI workflow (typecheck + tests).
- **Traceability table** in the README: requirement IDs mapped to IFU sections and tests.
- **Typed i18n dictionary:** a missing Spanish key fails the build.

React Native · Expo · TypeScript · Jest · GitHub Actions · i18n

LEFT OUT ON PURPOSE

- Real camera capture and image analysis
- Phone allow-list, storage, sync, auth
- Risk file and V&V records

Unofficial concept built from the public BRAINCheck Instructions for Use (ABC-IFU-001 v1.0, July 2022). Synthetic data only, simulated camera. Not a medical device and not clinical advice. Not affiliated with or endorsed by ABCDx; the network glyph is an original drawing, not their logo. The Spanish text is a draft pending native review.